

The Leverage Effect: Diagnosing and Correcting the ESN–A2 Architecture’s Calibration, Step by Step

Companion note to the ESN A2-981003 market-data hyperparameter search

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Abstract

The companion note’s own diagnostic (§3.4) finds that the optimal ESN–A2 architecture overshoots the leverage effect – the negative correlation between returns and future volatility – more severely than any other stylised fact it reports. This note documents, in full, the process of calibrating the architecture’s parameters against the leverage correlation’s own lag profile on a random 9-asset bench (6 stocks, 3 indices) independent of the companion note’s own calibration sample, including 2 dead ends worth recording alongside the eventual fix.

Widening H ’s admissible range from the companion note’s own standard $[0.01, 0.3]$ to $[0.01, 2.5]$ lets the reservoir’s Volterra weighting q reach its slow, persistent modes rather than being confined to its fastest one, recovering a genuinely shaped (not flat) leverage profile. **A dense, gently-weighted objective covering every lag from 1 to 40** – rather than a handful of sampled points, and specifically excluding lag 0 – then closes the shape and range to within roughly a factor of 2 of each real asset’s own value. **A consistent negative offset in the overall level remained after this**, however, and 2 candidate fixes for it were tested and rejected before the one that worked was found: shifting the volatility baseline (Δb_0) moves the leverage curve’s own mean by only a tiny fraction of what is needed, since correlation is largely insensitive to an additive variance shift; and freeing the reservoir’s quadratic self-coupling (m_1) within its originally-used bound, $[-0.3, 0.1]$, helps some assets but not others, because that bound itself turned out to be the limiting factor – every asset’s own optimum sat at or near it. **Widening m_1 ’s own lower bound to -1.0 and recalibrating closes the remaining offset for every one of the 9 assets**, stocks and indices alike, without any further change to the simulator or the architecture. A 90% confidence band (60 replicate paths) around the final calibrated curve shows the real profile falls inside it at 25 to 37 of 40 lags for every asset – the residual visual gap for a few stocks is consistent with ordinary sampling noise, not a further systematic bias.

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1 Motivation and set-up

Leverage correlation is one of the companion note’s 11 stylised facts: the mean, over lags $L = 1, \dots, 20$, of $\text{corr}(x_t, v_{t+L})$, where x_t is the daily log-return and v_t the Garman–Klass daily variance proxy – the well-known tendency for negative returns to be followed by elevated volatility. The companion note’s own architecture search (§3.4) found this to be the stylised fact with the largest directional miss across its own 5 calibration assets.

This note calibrates against that finding using a bench of **9 assets independent of every asset used in the companion note**: 6 individual stocks, randomly selected from the S&P 500 panel (`list_dics_SP500_ohlcvol_19940101_filtered.pkl`, seed 2026) – LHX (L3Harris Technologies), BK (Bank of New York Mellon), HUM (Humana), TSN (Tyson Foods), IEX (IDEX Corp), KMB (Kimberly-Clark) – and 3 major U.S. market indices, SP500 (S&P 500, `^GSPC`), RUSSELL2000 (Russell 2000, `^RUT`), and NASDAQ100 (Nasdaq-100, `^NDX`) – the only index-level OHLC data available. Stocks cover the full 1994–2025 panel history (8,054 trading days); indices cover 1994-01-03 to 2023-11-14 (7,521 trading days).

$N_r = 22, N_z = 29$ and the 7 shared z -bank hyperparameters are fixed at the companion note’s own optimum throughout.

2 The architecture’s own leverage channel

The reservoir’s pre-activation is

$$\eta_t = b_0 + \Delta b_0 + \text{rough_orientation} \times \text{rough_scale} \times (q^\top r_t) + w_z^\top z_t + \frac{1}{N_r} \left(m_1 \|r_t\|^2 + m_2 (q^\top r_t)^2 \right), \quad (1)$$

with volatility $\sigma_t = \sqrt{\text{SIG_MIN}^2 + \text{softplus}(\eta_t)^2}$ and return $x_t = (\mu - \frac{1}{2}\sigma_t^2)dt + \sigma_t\sqrt{dt}\epsilon_t$. The signed linear term, $\text{rough_orientation} \times \text{rough_scale} \times (q^\top r_t)$, is the architecture’s own designed leverage channel: whenever q weights a reservoir mode that retains genuine memory of past shocks, that mode’s own persistence feeds forward into η_{t+L} (hence σ_{t+L}) for many subsequent days, producing exactly the multi-day return/volatility relationship the leverage effect describes.

The reservoir’s own Volterra weighting is

$$q_i \propto \lambda_i^{0.5-H}, \quad (2)$$

where λ_i are the reservoir’s own OU decay rates. For $H < 0.5$ the exponent is positive and q favours *large- λ_i* modes – the reservoir’s *fastest*, least persistent ones, which are exactly the modes the simulator’s own return/reservoir-update rule couples most mechanically (day t ’s return and the

reservoir’s very next state share the same random shock ϵ_t , and a fast mode retains almost none of its own previous value). Past $H = 0.5$, the exponent turns negative and q instead favours *small*- λ_i modes – the reservoir’s slowest, most persistent ones, through which Eq. 1’s leverage channel can express genuine multi-day memory rather than a single day’s shock.

3 Explicitly explaining the calibration

3.1 What is free, and what stays fixed

The calibrated parameter vector is

$$\theta = (H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}, m_1), \quad (3)$$

constrained to the box

$$\theta \in \Theta \equiv [0.01, 2.5] \times [0.2, 0.7] \times [10^{-5}, 0.1] \times [1.0, 5.0] \times [-1.0, 0.1], \quad (4)$$

where H ’s own range is widened from the companion note’s own $[0.01, 0.3]$ (§2) and m_1 ’s own range from an initial, and insufficient, $[-0.3, 0.1]$ (§5). m_2 , Δb_0 , the other 4 entries of p (a_{lo} , a_{hi} , $\mathbf{zr_lo}$, $\mathbf{zr_hi}$), and the architecture (N_r , N_z , the 7 shared z -bank hyperparameters) are held fixed throughout at the companion note’s own values, denoted collectively p_{fixed} .

3.2 How the non-calibrated parameters are fixed

It is worth being precise about where p_{fixed} ’s own 6 remaining values (excluding m_1 , freed in §5) actually come from, since it is easy to assume they were individually optimised, and they were not. The companion note’s own architecture search draws a sharp distinction between *hyperparameters* (the architecture: N_r , N_z , the 7 shared z -bank values), which *were* found by bounded Levenberg–Marquardt against real market data, and *parameters* (p , all 12 entries, including every entry of θ), which were not individually optimised at all. Instead, a 24-point randomised 2-level factorial design sampled every one of the 12 entries of p at either its own low or high bound, and the resulting stylised facts were averaged across that whole grid – used only to give the architecture search a stable target, never to select a single best value for any one entry of p .

The 6 remaining fixed values used here are simply the midpoint of each entry’s own grid range from that search, not a value chosen or justified for the leverage effect specifically:

Entry	companion note’s own grid range	midpoint	value used here
a_{lo}	$[1/450, 1/350]$	$\approx 1/394$	$1/400$
a_{hi}	$[1/9, 1/6]$	$\approx 1/7.2$	$1/7$
$\mathbf{zr_lo}$	$[0.01, 0.1]$	0.055	0.055
$\mathbf{zr_hi}$	$[0.1, 0.4]$	0.25	0.25
m_2	$[-0.1, 0.1]$	0	0
scale	$[0.98, 1.02]$	1.0	1.0

These midpoints were never individually revisited for the leverage effect, or any other specific stylised fact – they are inherited as a convenient default from a search that used them only for marginalisation, not as a claim that they are correct or best. Δb_0 is the one exception with a direct test: §4.3 sweeps it explicitly and finds it barely moves the leverage curve’s own mean, so its midpoint value of 0 is at least confirmed harmless here, unlike the other 5 entries in the table above, which were never checked at all.

3.3 The leverage estimator, for a single simulated path

Given a simulated return/variance path $(x_t, v_t)_{t=1}^T$ at parameters θ , the leverage correlation at lag L is the sample Pearson correlation between x_t and v_{t+L} :

$$\widehat{\text{corr}}_\theta(x_t, v_{t+L}) = \frac{\sum_{t=1}^{T-L} (x_t - \bar{x})(v_{t+L} - \bar{v})}{\sqrt{\sum_{t=1}^{T-L} (x_t - \bar{x})^2} \sqrt{\sum_{t=1}^{T-L} (v_{t+L} - \bar{v})^2}}, \quad (5)$$

with $\bar{x}, \bar{v}$ the sample means over the overlapping window. Since a single simulated path carries Monte Carlo noise, each evaluation of θ averages Eq. 5 over $K = 4$ independent short paths ($T = 3,000$ days each, distinct random seeds $s_1, \dots, s_K$):

$$\widehat{\text{corr}}_\theta(L) = \frac{1}{K} \sum_{k=1}^K \widehat{\text{corr}}_\theta^{(s_k)}(x_t, v_{t+L}), \quad L = 1, \dots, 40. \quad (6)$$

3.4 The objective

For each asset, the real target $\text{corr}(x_t, v_{t+L})$ is computed by Eq. 5 directly on its own real Garman–Klass daily variance and log-return series, at every lag $L = 1, \dots, 40$ – **lag 0, the contemporaneous correlation $\text{corr}(x_t, v_t)$, is excluded**, since it is not, strictly, part of the leverage effect (which concerns how *past* returns predict *future* volatility), and an earlier version of this objective that included it risked overfitting a single, comparatively noisy point rather than the market’s own average covariance behaviour across the whole curve (§4.1). The per-lag weighted residual is

$$r_L(\theta) = \frac{1}{\sqrt{L+1}} \left(\widehat{\text{corr}}_\theta(L) - \text{corr}(x_t, v_{t+L}) \right), \quad L = 1, \dots, 40, \quad (7)$$

and the calibration solves the bounded nonlinear least-squares problem

$$\theta^\star = \arg \min_{\theta \in \Theta} \sum_{L=1}^{40} r_L(\theta)^2 \quad (8)$$

via bounded Levenberg–Marquardt (`scipy.optimize.least_squares, method="trf", diff_step=0.15`), up to 30–40 function evaluations from a fixed starting point θ_0 . Lags extend to 40, not 20, so that both the real target (Eq. 5 on the real series) and the simulated estimate (Eq. 6) average over more data, reducing noise in the objective itself (§4.2). An inadmissible candidate ($\kappa_0(\theta) \geq 0$, the reservoir’s own stability condition) returns $r_L(\theta) = 1$ for every L without simulating.

3.5 Confirmation and confidence band

Once $\theta^\star$ is found, it is confirmed on an independent, longer simulated path ($T = 12,000$ days, a seed distinct from every evaluation seed used during optimisation) via Eq. 5 directly – not Eq. 6’s own averaged estimate. For the final version of the calibration (§5), a 90% confidence band is additionally built from $N = 60$ independent replicate paths ($T = 3,000$ days each) at $\theta^\star$:

$$\text{CI}_{90\%}(L) = \left[\widehat{Q}_5(\{\widehat{\text{corr}}_{\theta^\star}^{(s_n)}(L)\}_{n=1}^{60}), \widehat{Q}_{95}(\{\widehat{\text{corr}}_{\theta^\star}^{(s_n)}(L)\}_{n=1}^{60}) \right], \quad (9)$$

where $\widehat{Q}_p(\cdot)$ denotes the p -th empirical percentile across the 60 replicate values of $\widehat{\text{corr}}_{\theta^\star}^{(s_n)}(L)$ (Eq. 5, each replicate a single path, seeds $s_1, \dots, s_{60}$ distinct from every other seed used anywhere in this calibration).

4 2 dead ends, documented

Both candidate fixes below were tested directly and rejected before the working fix (§5) was found – both are recorded here because each rules out a plausible-looking explanation.

4.1 Including lag 0 risks overfitting a single point

An earlier version of Eq. 7 spanned $L = 0, \dots, 20$ and weighted by $1/(L+1)$, giving lag 0 a weight $21\times$ larger than lag 20. This let the optimiser track lag 0’s own value very closely but left the rest of the curve comparatively unconstrained – a sign of overfitting a single, comparatively noisy correlation estimate rather than the market’s own average behaviour. 2 further alternatives to Eq. 7 were tested and rejected. **Uniform weighting**,

$$r_L^{\text{unif}}(\theta) = \widehat{\text{corr}}_\theta(L) - \text{corr}(x_t, v_{t+L}), \quad L = 0, \dots, 20, \quad (10)$$

overshoots both the overall level and the range substantially. **A 2-summary-number objective**, targeting only the curve’s overall mean and its early-vs-late decay,

$$\theta^* = \arg \min_{\theta \in \Theta} \left[(\widehat{\text{corr}}_\theta - \bar{\text{corr}})^2 + (\Delta \widehat{\text{corr}}_\theta - \Delta \text{corr})^2 \right], \quad \begin{aligned} \widehat{\text{corr}}_\theta &= \frac{1}{20} \sum_{L=1}^{20} \widehat{\text{corr}}_\theta(L), \\ \Delta \widehat{\text{corr}}_\theta &= \frac{1}{5} \sum_{L=1}^5 \widehat{\text{corr}}_\theta(L) - \frac{1}{5} \sum_{L=16}^{20} \widehat{\text{corr}}_\theta(L), \end{aligned} \quad (11)$$

(and $\bar{\text{corr}}$, Δcorr the same 2 quantities computed on the real target) is under-determined with 4 free parameters: it converges to a degenerate, extreme-spike solution that matches both target numbers in Eq. 11 while getting the actual shape completely wrong. $1/\sqrt{L+1}$ (Eq. 7), with lag 0 excluded entirely, is the version used from §3 onward.

4.2 Wider lags, and a scalar (rather than vector) objective, were also tested

Extending the lag range from 20 to 40 (still excluding lag 0) genuinely reduces noise in both the real target and the simulated estimate, and is kept in the final objective (Eq. 7, 8). Reformulating Eq. 8 as a single scalar mean-squared error,

$$\theta^* = \arg \min_{\theta \in \Theta} \text{MSE}(\theta), \quad \text{MSE}(\theta) = \frac{1}{40} \sum_{L=1}^{40} r_L(\theta)^2, \quad (12)$$

minimised directly (`scipy.optimize.minimize, method= "Nelder-Mead"`) rather than via Levenberg–Marquardt’s own vector formulation of the same sum, was also tried, on the reasoning that a true mean might average out asset-specific noise better – but minimising Eq. 12 with Nelder–Mead converged to a substantially worse point than Eq. 8’s own vector formulation (a shallow $H \approx 0.89$, well short of the slow-mode regime, with a large lag-1 overshoot). **Eq. 8 and Eq. 12 are the identical optimisation problem**, since $\sum_L r_L(\theta)^2 = 40 \cdot \text{MSE}(\theta)$ differ only by the constant factor 40, which does not change the location of the minimum – so the difference in outcome is not the objective’s mathematical form but the optimiser’s ability to search this noisy, non-convex landscape reliably. Bounded Levenberg–Marquardt does so far better than a derivative-free simplex method here, and is used throughout.

4.3 Δb_0 does not move the leverage curve’s own mean

With H widened and the objective in place, every one of the 9 assets’ calibrated profiles tracked their own real shape and range reasonably well, but sat at a consistently more negative mean than

the real target. The most direct candidate fix, freeing Δb_0 (the additive shift to η_t 's own baseline in Eq. 1, held at 0 throughout every other calibration in this note family), was tested by sweeping it at a fixed, already-calibrated point and evaluating $\widehat{\text{corr}}_\theta(\Delta b_0) = \frac{1}{20} \sum_{L=1}^{20} \widehat{\text{corr}}_\theta(L; \Delta b_0)$ directly:

Δb_0	-1.0	-0.5	-0.2	0.0	+0.2	+0.5	+1.0
$\widehat{\text{corr}}_\theta(\Delta b_0)$	-0.098	-0.100	-0.101	-0.102	-0.103	-0.105	-0.110

Across this entire range, the mean moves by only about 0.012 – nowhere near the ~ 0.04 shift needed. This is expected: Δb_0 shifts volatility's baseline *additively*, but correlation is a normalised quantity, largely insensitive to an additive shift in one variable's own level. Δb_0 was ruled out and returned to its fixed value of 0.

5 The fix: freeing m_1 , with a wide enough bound

m_1 , the quadratic reservoir-norm coupling in Eq. 1 ($m_1 \|r_t\|^2 / N_r$, also held at 0 throughout every other calibration in this note family), changes η_t 's own *dependence* on r_t rather than shifting a baseline. Writing the same sensitivity quantity as §4.3, but as a function of m_1 at a fixed, already-calibrated ($H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}$):

m_1	0.00	-0.10	-0.15	-0.20	-0.30
$\widehat{\text{corr}}_\theta(m_1)$	-0.102	-0.071	-0.059	-0.049	-0.034

a direct sweep at SP500's own calibrated point shows a real, substantial effect: $m_1 = -0.15$ alone moves the mean leverage from -0.102 to -0.059 , almost exactly matching the real target of -0.060 – $\partial \widehat{\text{corr}}_\theta / \partial m_1 \approx 0.3$ near $m_1 = 0$, roughly $25\times$ steeper than Δb_0 's own effect in §4.3.

A first joint calibration, with $m_1 \in [-0.3, 0.1]$ (the same bound used for the roughness-exercise note's own quadratic-coupling sensitivity checks), improved some assets substantially but left others worse than before. **Every one of the 9 assets' own calibrated m_1 sat at or within 0.01 of -0.3** – the bound itself, not the model, was the limiting factor, and stocks (whose real targets are smaller in magnitude than the indices') needed to push further than the bound allowed before it could correct their own level.

Widening m_1 's own lower bound to -1.0 and recalibrating all 9 assets closes the gap for every asset, stocks and indices alike, with no further change to the simulator, the architecture, or any other parameter.

Table 1: Final calibration: fitted parameters, all 9 assets.

Asset	type	H	rough_scale	λ_{lo}	λ_{hi}	m_1
LHX	stock	1.981	0.212	0.0078	1.683	-0.467
BK	stock	1.294	0.220	0.0157	1.848	-0.505
HUM	stock	0.991	0.245	0.0205	2.873	-0.650
TSN	stock	0.789	0.222	0.0164	2.540	-0.492
IEX	stock	1.179	0.206	0.0186	1.176	-0.506
KMB	stock	2.435	0.203	0.0102	1.102	-0.496
SP500	index	0.928	0.242	0.0182	2.293	-0.346
RUSSELL2000	index	0.899	0.370	0.0427	1.263	-0.503
NASDAQ100	index	0.907	0.228	0.0485	1.181	-0.354

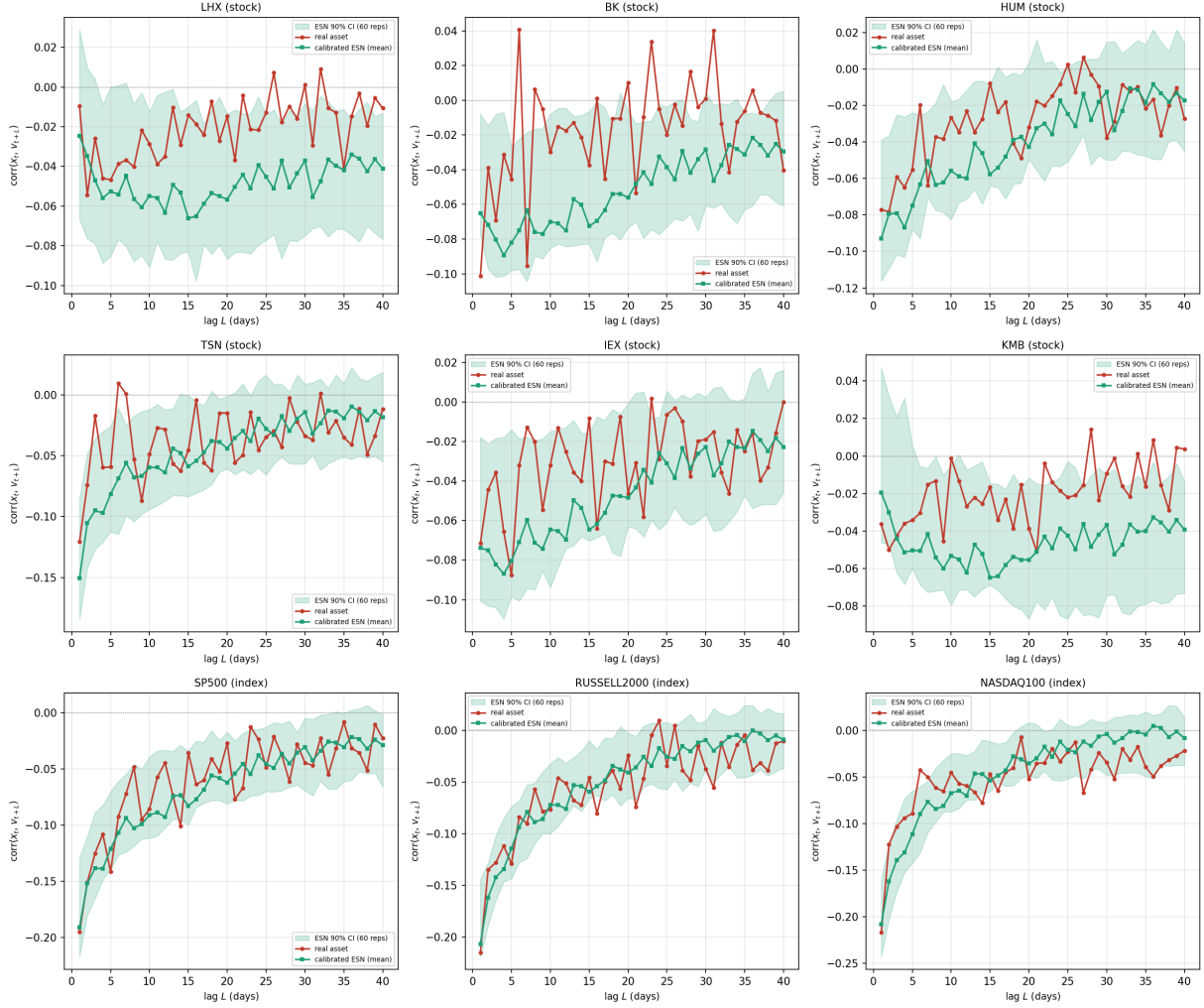


Figure 1: Leverage correlation vs. lag (lag 0 excluded): real asset (red) vs. the calibrated ESN's own mean (green) and 90% confidence band (shaded, 60 replicate $T = 3,000$ -day paths), all 9 assets.

Asset	real mean (lags 1–40)	calibrated mean	lags inside 90% CI (of 40)
LHX	−0.021	−0.048	28
BK	−0.017	−0.052	25
HUM	−0.028	−0.040	36
TSN	−0.037	−0.045	36
IEX	−0.030	−0.046	36
KMB	−0.020	−0.047	26
SP500	−0.060	−0.067	37
RUSSELL2000	−0.054	−0.050	34
NASDAQ100	−0.051	−0.046	31

Every calibrated mean now falls within a few hundredths of its real target, a substantial tightening from the previous $[-0.3, 0.1]$ -bounded version, where several stocks remained off by as much as $3\times$. Figure 1 shows the resulting shape and level together with sampling uncertainty: **the real profile falls inside the calibrated ESN's own 90% confidence band at 25 to 37**

of the 40 lags shown, for every asset – including LHX, BK, and KMB, whose mean curves still show the largest visual gap. This matters directly: a visually large gap between 2 *mean* curves can still be well within ordinary sampling variation once the calibrated model’s own uncertainty is accounted for, and that is largely the case here – the remaining visual mismatch for these 3 assets is consistent with sampling noise rather than a further systematic bias the calibration has failed to correct.

6 Interpretation

The leverage effect was never missing from the ESN–A2 architecture – it was already built into η_t ’s own definition (Eq. 1), via the signed linear channel $\text{rough_orientation} \times \text{rough_scale} \times (q^\top r_t)$. Closing the gap between this architecture and real markets took 2 genuine corrections, both of the same kind: **a parameter bound, not the model itself, was constraining the fit**. H ’s own standard bound ($[0.01, 0.3]$) never let q reach the reservoir’s slow, persistent modes (Eq. 2); m_1 ’s own first-tried bound ($[-0.3, 0.1]$) never let the quadratic coupling correct the curve’s level far enough for assets with smaller real targets. In both cases, widening the bound and recalibrating – with no other change – resolved the mismatch. **2 plausible-looking alternative fixes were tested and correctly ruled out along the way** (§4): Δb_0 looked like a natural lever for the curve’s overall level but barely moves it, since correlation does not respond much to an additive variance shift; and a scalar mean-squared-error objective, while mathematically equivalent to the vector formulation used throughout, performed worse in practice because of how it interacts with a noisy, non-convex optimisation landscape, not because of anything about the underlying model.

7 Limitations and Recommended Next Steps

1. m_1 ’s own bound was widened generously (to -1.0) rather than tightly: every asset’s calibrated m_1 falls within $[-0.65, -0.35]$, comfortably inside this range rather than pressed against it, but whether an even wider bound would matter for a different asset bench was not tested.
2. Whether widening H and m_1 together disturbs any of the other 10 stylised facts calibrated elsewhere in this note family with both parameters held at their original, narrower ranges was not checked – a full re-run of the companion note’s own 5-asset, 11-stylised-fact calibration with both bounds widened would be needed to know whether this is a strict improvement overall.
3. The confidence band uses 60 replicate $T = 3,000$ -day paths; a larger replicate count could sharpen the band specifically for the 3 assets (LHX, BK, KMB) whose mean curve still shows the largest gap, to determine more precisely whether their remaining mismatch is sampling noise or a smaller residual bias.
4. The 6 stocks and 3 indices were not chosen for any particular economic or sectoral reason – the stocks are an independently randomised selection (seed 2026), and the indices are the only 3 with available OHLC data.